Introduction to R

Session 05: Data Wrangling

Álvaro Pérez

Instituto Tecnológico Autónomo de México

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Overview of Today's Session

Main dplyr verbs

Additional dplyr one-table verbs

Window functions

Dplyr verbs

select()

- Select columns by name or number.
- You can select columns by number, which is useful when the column names are long or complicated.
- You can use a minus symbol to unselect columns, leaving all of the other columns.

Select helpers

- starts_with(): Select columns that start with a character string.
- ends_with(): Select columns that end with a character string.
- contains(): Select columns that contain a character string.
- num_range(): Select columns with a name that matches the pattern prefix.

Dplyr verbs

filter()

- Select rows by matching column criteria.
- Remember to use $==$ and not $=$ to check if two things are equivalent. A single $=$ assigns the righthand value to the lefthand variable and (usually) evaluates to TRUE.
- You can use the symbols &, $|$, and ! to mean "and," "or," and "not." You can also use other operators to make equations.

Match operator (%in%): The match operator is useful here for testing if a column value is in a list.
Dates: you can use the year() function to return just the year from the date column and then select only data collected in 2010.

Dplyr verbs

- **arrange ()**: sort your dataset
- **mutate ()**: add new columns. You can add more than one column in the same mutate function, just separate the columns with a comma.
- **summarise ()**: create summary statistics for the dataset. Some common ones are: mean(), sd(), n(), sum(), and quantile().
- **group_by ()**: create subsets of the data. You can use this to create summaries, like the mean value for all of your experimental groups.

Additional dplyr verbs

- **rename()**: rename columns
- **distinct()**: get rid of exactly duplicate rows. This can be helpful if, for example, you are merging data from multiple computers and some of the data got copied from one computer to another, creating duplicate rows.
- **count ()**: a quick shortcut for the common combination of group_by() and summarise() used to count the number of rows per group.

Window functions

- **Offset functions**: The function lag() gives a previous row's value. It defaults to 1 row back, but you can change that with the n argument. The function lead() gives values ahead of the current row.
- **Cumulative aggregates**: cumsum(), cummin(), and cummax() are base R functions for calculating cumulative means, minimums, and maximums. The dplyr package introduces cumany() and cumall(), which return TRUE if any or all of the previous values meet their criteria.